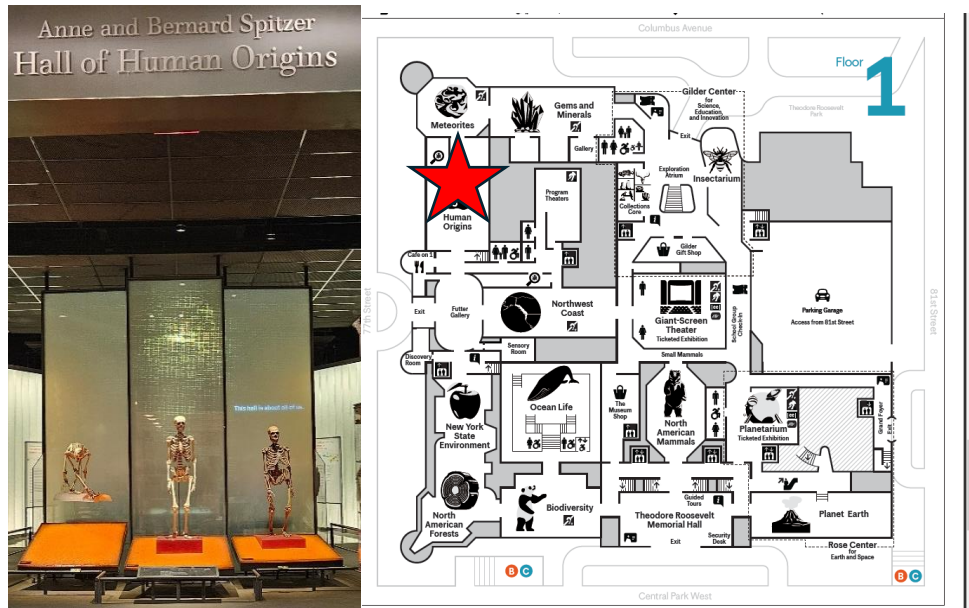


First, you will travel to the Hall of Human Origins :



The first photo opportunity is below. Anytime is a good time for family reunion! Say hi to your cousins from 10^4 and 10^{12} years ago!



CLUE set #1 Hall of Human Origins:

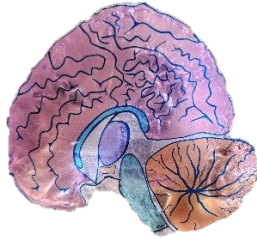
- As you will observe while walking through this area, the evolution of our bodies from primates to *Homo sapiens* was subtle and took a very long time.
- There are a lot of skulls around showing similar features and the evolution of those features over time, but what about the brains that they protected?
- While looking through this area find the place that illustrates how our brains have changed over time from 3 million years ago to today.
- Questions for exhibit 1:
 1. Who were the four predecessors described in the exhibit excluding *Homo sapiens*? How long was each walking the earth?
 2. What ability does our brain possess that sets us apart from all other species including our ancestors?
 3. How big was the brain of our ancestor who lived 3 million years ago?
- Nearby you will find another exhibit showing how our brain to body ratio stacks up against some other common species.
- Questions for exhibit 2:
 1. Identify the species and brain's percent of body size for each of the images below.



d)



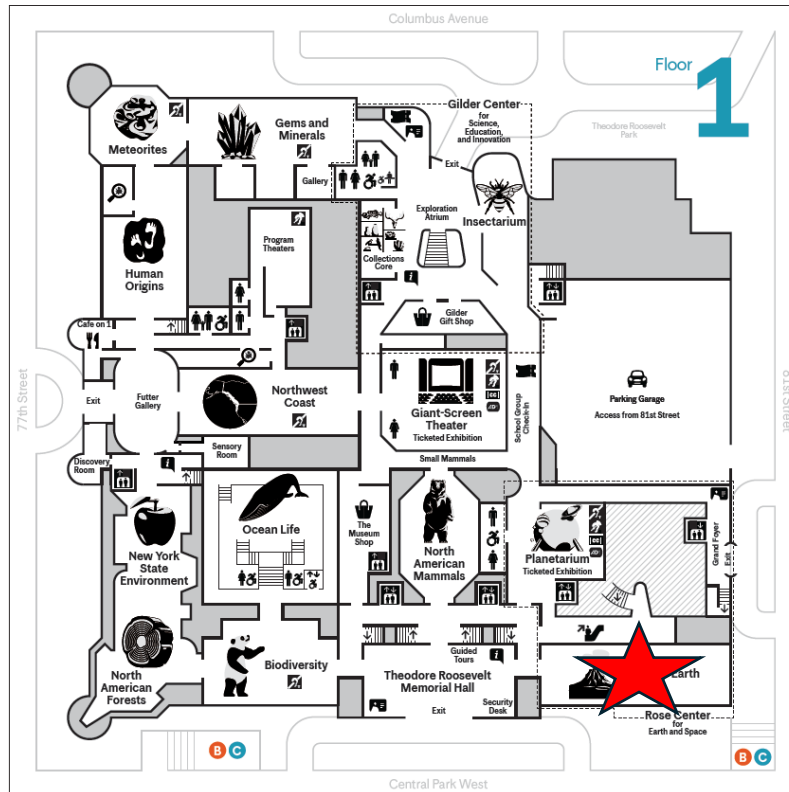
e)



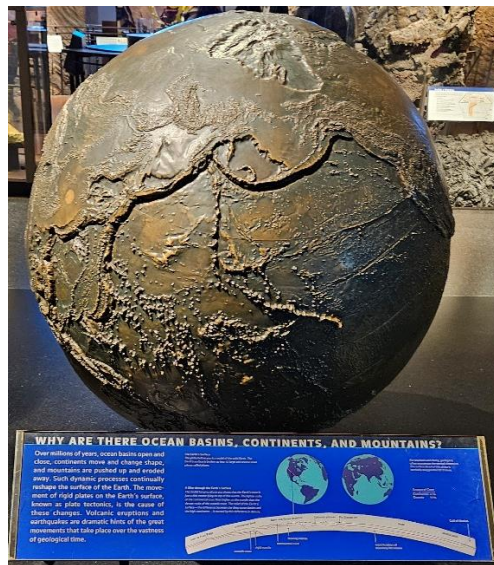
2. Using your knowledge of brain structure identify the pink region in brains of parts **b – e**.
3. Using your knowledge of brain function, name two functions that the brain of species **e** would likely be better at when compared to species **b**, based on the size and structure of the pink region.
4. An elephant has a brain to body ratio of 1:150 kg. What is the weight of an elephant's brain whose body weight is 900 kg? Answer in kg. Don't use a calculator!

On To The Next Exhibit!!

Gottesman Hall of PLANET EARTH

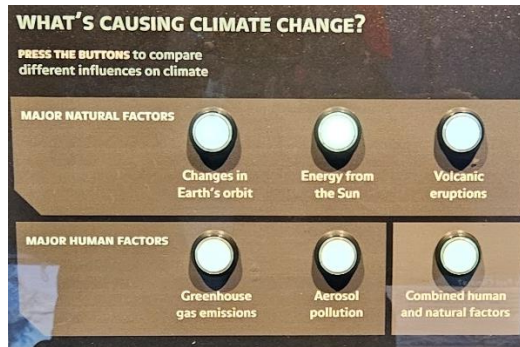


Here you will find the second photo opportunity. Take a pic with the bronze model of the third planet from the sun that's the first in our hearts!



CLUE set #2: The Hall of Planet Earth:

- The final module of Frontiers is Climate and Us, so take the opportunity to learn a bit about our climate and the factors that affect it.
- Make your way to the big screen to look at some data on climate change on our planet.
- Once you arrive at the screen look for a set of buttons that look like this:

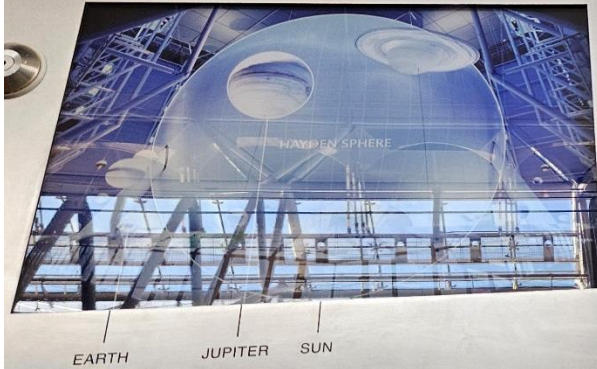


1.

- Questions for exhibit 1:
 1. Which of the major natural factors have the most impact on climate change?
 2. Which of the major human factors has the most impact on climate change?
- Look up! Is that a tornado? Nope, but it is a disaster in the making 😞
 3. What is the important takeaway of the animation of global temperature over time? (It plays every few minutes)
- Investigate the panel under the screen.
 4. What were the yearly odds of an extreme heat wave in Europe in the early 1900s vs 2015 ?

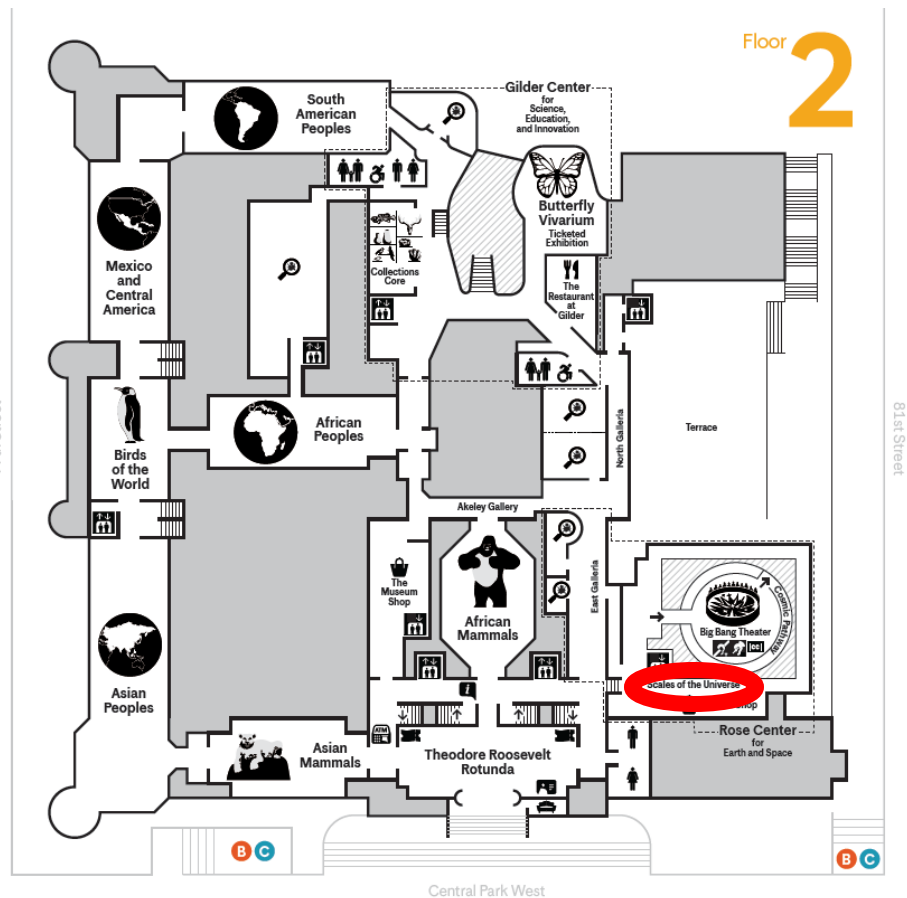
On To The Next Exhibit!!

SIZE SCALES of the universe



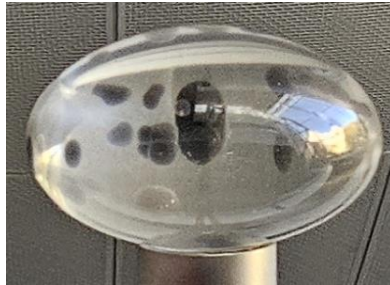
This exhibit explores the known range of size scales in the cosmos, from the observable universe at the very largest, down to the nucleus of the atom at the very smallest.

The Hayden Sphere in front of you serves as a scale reference for exploring this entire range of sizes. For example, if the Hayden Sphere — 26.5 meters (87 feet) in diameter — represents the size of the Sun, then Jupiter stretches 2.7 meters (9 feet) across, while Earth is a mere 24 centimeters (9.5 inches) in diameter.



Clue Set # 3

- The second module of Frontiers was Physics and Reality. During this module Dr. Green introduced us to the reality cube which informed us of the range in scales of the universe.
- Similarly, this next exhibit gives us the chance to see the relative size of different celestial bodies compared to the Hayden Sphere.
- Questions for exhibit 1:
 1. Given the below images identify the approximate length of the celestial object and its name:



■



■



■

Now head down to the lower level to learn about black holes.

Once you reach the lower level, locate this theater:



Have a seat and watch the documentary. It plays every few minutes. Answer the following questions about black holes.

2. What happens when a star runs out of fuel?
3. What year was the first evidence of a black hole's existence discovered and what was the black hole's name?
4. How hot does an accretion disk get?
5. What is gravitational lensing and how does it relate to general relativity? (This won't be answered in video. Do some internet research and come up with a good answer.)



Thank you for completing this little scavenger hunt! I hope you had a good time exploring the museum and I can't wait to hear about your experiences in your reflection, if you decide to write one.

What To Turn In

For the Scavenger hunt assignment, you can either create a separate document with your answers to the questions or fill in the answers on this document. The photo that will act as your attendance should be attached to the document. Make sure you submit everything on courseworks as one document.